Tutorial 9: System of First-order Linear DEs (Part 2)

1. Let

$$\boldsymbol{A} = \begin{pmatrix} 1 & -1 \\ 1 & 3 \end{pmatrix}.$$

Determine if $\boldsymbol{A}$ is defective or non-defective. Find the general solution of the system $\boldsymbol{x}'(t) = \boldsymbol{A}\boldsymbol{x}(t)$.

Solution: We first find the eigenvalues and eigenvectors of $\boldsymbol{A}$:

$$0 = \det(\boldsymbol{A} - \lambda \boldsymbol{I}_2) = \begin{vmatrix} 1-\lambda & -1 \\ 1 & 3-\lambda \end{vmatrix} = (1-\lambda)(3-\lambda) + 1 = \lambda^2 - 4\lambda + 4 = (\lambda - 2)^2.$$

Thus, the eigenvalue: $\lambda = 2$ (repeated).

We look for the corresponding eigenvectors:

$$(\boldsymbol{A} - \lambda \boldsymbol{I}_2)\boldsymbol{v} = \boldsymbol{0} \quad \Rightarrow \quad \begin{pmatrix} -1 & -1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

So the eigenvectors and corresponding solution are

$$\boldsymbol{v} = r \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \quad r \neq 0, \quad \boldsymbol{x}_1(t) = e^{2t} \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

Therefore, the matrix is defective.

We look for the second LI solution of the form

$$\boldsymbol{x}_2(t) = (\boldsymbol{v}t + \boldsymbol{w})e^{2t},$$

where $\boldsymbol{v}$ and $\boldsymbol{w}$ satisfy

$$(\boldsymbol{A} - \lambda \boldsymbol{I}_2)\boldsymbol{v} = \boldsymbol{0}, \quad (\boldsymbol{A} - \lambda \boldsymbol{I}_2)\boldsymbol{w} = \boldsymbol{v} \quad \Rightarrow \quad (\boldsymbol{A} - \lambda \boldsymbol{I}_2)^2\boldsymbol{w} = \boldsymbol{0}.$$

We solve $\boldsymbol{w}$ and then obtain $\boldsymbol{v}$ via $\boldsymbol{v} = (\boldsymbol{A} - \lambda \boldsymbol{I}_2)\boldsymbol{w}$. More precisely, we have

$$(\boldsymbol{A} - \lambda \boldsymbol{I}_2)^2\boldsymbol{w} = \boldsymbol{0} \quad \Rightarrow \quad \begin{pmatrix} -1 & -1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} -1 & -1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} w_1 \\ w_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$
$$\Rightarrow \quad \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} w_1 \\ w_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix},$$

so $\boldsymbol{w}$ can be any nonzero vector. We take $\boldsymbol{w} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and compute

$$\boldsymbol{v} = (\boldsymbol{A} - \lambda \boldsymbol{I}_2)\boldsymbol{w} = \begin{pmatrix} -1 & -1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}.$$

Thus, the second LI solution is

$$\boldsymbol{x}_2(t) = te^{2t} \begin{pmatrix} -1 \\ 1 \end{pmatrix} + e^{2t} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1-t \\ t \end{pmatrix} e^{2t}.$$

Then the general solution is

$$\boldsymbol{x}(t) = c_1 \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{2t} + c_2 \begin{pmatrix} 1-t \\ t \end{pmatrix} e^{2t}.$$

2. Let

$$\boldsymbol{A} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 4 \end{pmatrix}.$$

Determine if $\boldsymbol{A}$ is defective or non-defective. Find the general solution of the system $\boldsymbol{x}'(t) = \boldsymbol{A}\boldsymbol{x}(t)$.

Solution: We first find the eigenvalues and eigenvectors of $\boldsymbol{A}$:

$$0 = \det(\boldsymbol{A} - \lambda \boldsymbol{I}_3) = \begin{vmatrix} 1-\lambda & 1 & 0 \\ 0 & 1-\lambda & 1 \\ 0 & 0 & 4-\lambda \end{vmatrix} = (1-\lambda)^2(4-\lambda).$$

Thus, the eigenvalues are $\lambda_1 = 4$ and $\lambda_2 = 1$ (repeated).

We look for the corresponding eigenvectors. For $\lambda_1 = 4$, we solve

$$(\boldsymbol{A} - \lambda_1 \boldsymbol{I}_3)\boldsymbol{v} = \boldsymbol{0} \quad \Rightarrow \quad \begin{pmatrix} -3 & 1 & 0 \\ 0 & -3 & 1 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

So the eigenvectors and corresponding solution are

$$\boldsymbol{v} = r \begin{pmatrix} 1 \\ 3 \\ 9 \end{pmatrix}, \quad r \neq 0, \quad \boldsymbol{x}_1(t) = e^{4t} \begin{pmatrix} 1 \\ 3 \\ 9 \end{pmatrix}.$$

For $\lambda_2 = 1$, we solve

$$(\boldsymbol{A} - \lambda_2 \boldsymbol{I}_3)\boldsymbol{v} = \boldsymbol{0} \quad \Rightarrow \quad \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 3 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

So the eigenvectors and corresponding solution are

$$\boldsymbol{v} = s \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad s \neq 0, \quad \boldsymbol{x}_2(t) = e^t \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}.$$

Therefore, there are only two LI eigenvectors in total. The matrix is defective.

We seek the third LI solution in the form

$$\boldsymbol{x}_3(t) = (\boldsymbol{v}t + \boldsymbol{w})e^t,$$

where $\boldsymbol{v}$ and $\boldsymbol{w}$ satisfy

$$(\boldsymbol{A} - \boldsymbol{I}_3)\boldsymbol{v} = \boldsymbol{0}, \quad (\boldsymbol{A} - \boldsymbol{I}_3)\boldsymbol{w} = \boldsymbol{v} \quad \Rightarrow \quad (\boldsymbol{A} - \boldsymbol{I}_3)^2\boldsymbol{w} = \boldsymbol{0}.$$

We solve $\boldsymbol{w}$ and then obtain $\boldsymbol{v}$ via $\boldsymbol{v} = (\boldsymbol{A} - \boldsymbol{I}_3)\boldsymbol{w}$. More precisely, we have

$$\begin{aligned} (\boldsymbol{A} - \boldsymbol{I}_3)^2\boldsymbol{w} = \boldsymbol{0} \quad &\Rightarrow \quad \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 3 \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 3 \end{pmatrix} \begin{pmatrix} w_1 \\ w_2 \\ w_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \\ &\Rightarrow \quad \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 3 \\ 0 & 0 & 9 \end{pmatrix} \begin{pmatrix} w_1 \\ w_2 \\ w_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \end{aligned}$$

so $w_3 = 0$ and w_1, w_2 are free to choose. We take $\boldsymbol{w} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$ and compute

$$\boldsymbol{v} = (\boldsymbol{A} - \boldsymbol{I}_3)\boldsymbol{w} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 3 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}.$$

Therefore,

$$\boldsymbol{x}_3(t) = (\boldsymbol{v}t + \boldsymbol{w})e^t = e^t \begin{pmatrix} t \\ 1 \\ 0 \end{pmatrix}.$$

Finally, the general solution is

$$\boldsymbol{x}(t) = c_1 e^{4t} \begin{pmatrix} 1 \\ 3 \\ 9 \end{pmatrix} + c_2 e^t \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + c_3 e^t \begin{pmatrix} t \\ 1 \\ 0 \end{pmatrix}.$$

3. Let

$$\begin{pmatrix} -2 & 0 & 0 \\ 1 & -3 & -1 \\ -1 & 1 & -1 \end{pmatrix}$$

Determine if $\boldsymbol{A}$ is defective or non-defective. Find the general solution of the system $\boldsymbol{x}'(t) = \boldsymbol{A}\boldsymbol{x}(t)$.

Solution: We first find the eigenvalues and eigenvectors of $\boldsymbol{A}$:

$$\begin{aligned} 0 = \det(\boldsymbol{A} - \lambda \boldsymbol{I}_3) &= \begin{vmatrix} -2-\lambda & 0 & 0 \\ 1 & -3-\lambda & -1 \\ -1 & 1 & -1-\lambda \end{vmatrix} \\ &= -(\lambda+2)((\lambda+1)(\lambda+3)+1) = -(\lambda+2)^3. \end{aligned}$$

Thus, it has one eigenvalue $\lambda = -2$ with a multiplicity 3.

We now look for the corresponding eigenvectors and solve

$$(\boldsymbol{A} + 2\boldsymbol{I}_3)\boldsymbol{v} = \boldsymbol{0} \quad \Rightarrow \quad \begin{pmatrix} 0 & 0 & 0 \\ 1 & -1 & -1 \\ -1 & 1 & 1 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

It is equivalent to $v_1 - v_2 - v_3 = 0$ and set the free variables: $v_2 = r$ and $v_3 = s$. The eigenvectors are

$$\boldsymbol{v} = r \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + s \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \tag{1}$$

so we get two LI solutions

$$\boldsymbol{x}_1(t) = e^{-2t} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \quad \boldsymbol{x}_2(t) = e^{-2t} \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}.$$

The dimension of the eigen-space is 2, so we have two LI eigenvectors. The matrix $\boldsymbol{A}$ is defective.

We seek the third LI solution in the form

$$\boldsymbol{x}_3(t) = (\boldsymbol{v}t + \boldsymbol{w})e^{-2t},$$

where $\boldsymbol{v}$ and $\boldsymbol{w}$ satisfy

$$(\boldsymbol{A} + 2\boldsymbol{I}_3)\boldsymbol{v} = \boldsymbol{0}, \quad (\boldsymbol{A} + 2\boldsymbol{I}_3)\boldsymbol{w} = \boldsymbol{v} \quad \Rightarrow \quad (\boldsymbol{A} + 2\boldsymbol{I}_3)^2\boldsymbol{w} = \boldsymbol{0}.$$

We solve $\boldsymbol{w}$ and then obtain $\boldsymbol{v}$ via $\boldsymbol{v} = (\boldsymbol{A} + 2\boldsymbol{I}_3)\boldsymbol{w}$. More precisely, we have

$$(\boldsymbol{A} + 2\boldsymbol{I}_3)^2\boldsymbol{w} = \boldsymbol{0} \quad \Rightarrow \quad \begin{pmatrix} 0 & 0 & 0 \\ 1 & -1 & -1 \\ -1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 \\ 1 & -1 & -1 \\ -1 & 1 & 1 \end{pmatrix} \begin{pmatrix} w_1 \\ w_2 \\ w_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow \quad \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} w_1 \\ w_2 \\ w_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix},$$

so $\boldsymbol{w}$ can be any nonzero vector. We take $\boldsymbol{w} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$ and compute

$$\boldsymbol{v} = (\boldsymbol{A} + 2\boldsymbol{I}_3)\boldsymbol{w} = \begin{pmatrix} 0 & 0 & 0 \\ 1 & -1 & -1 \\ -1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix}.$$

Note that if we take $r = -1$ and $s = 1$ in (1), we get this vector, so it is in the eigen-space. Therefore,

$$\boldsymbol{x}_3(t) = (\boldsymbol{v}t + \boldsymbol{w})e^{-2t} = e^{-2t} \begin{pmatrix} 0 \\ -t \\ 1+t \end{pmatrix}.$$

Finally, the general solution is

$$\boldsymbol{x}(t) = c_1 e^{-2t} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + c_2 e^{-2t} \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + c_3 e^{-2t} \begin{pmatrix} 0 \\ -t \\ 1+t \end{pmatrix}.$$

4. Solve the nonhomogeneous system

$$\boldsymbol{x}'(t) = \boldsymbol{A}\boldsymbol{x}(t) + \boldsymbol{F}(t),$$

where

$$\boldsymbol{A} = \begin{pmatrix} 0 & 1 \\ -9 & 6 \end{pmatrix}, \quad \boldsymbol{F}(t) = \begin{pmatrix} 0 \\ t \end{pmatrix}.$$

Solution: We first find the eigenvalues and eigenvectors of $\boldsymbol{A}$. Solve

$$0 = \det(\boldsymbol{A} - \lambda\boldsymbol{I}_2) = \begin{vmatrix} -\lambda & 1 \\ -9 & 6-\lambda \end{vmatrix} = \lambda(\lambda - 6) + 9 = (\lambda - 3)^2.$$

Thus, the eigenvalue is $\lambda = 3$ (repeated).

We look for the corresponding eigenvectors:

$$(\boldsymbol{A} - 3\boldsymbol{I}_2)\boldsymbol{v} = \boldsymbol{0} \quad \Rightarrow \quad \begin{pmatrix} -3 & 1 \\ -9 & 3 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

So the eigenvectors and corresponding solution are

$$\boldsymbol{v} = r \begin{pmatrix} 1 \\ 3 \end{pmatrix}, \quad r \neq 0, \quad \boldsymbol{x}_1(t) = e^{3t} \begin{pmatrix} 1 \\ 3 \end{pmatrix}.$$

Therefore, the matrix is defective.

We look for the second LI solution of the form

$$\boldsymbol{x}_2(t) = (\boldsymbol{v}t + \boldsymbol{w})e^{3t},$$

where $\boldsymbol{v}$ and $\boldsymbol{w}$ satisfy

$$(\boldsymbol{A} - \lambda\boldsymbol{I}_2)\boldsymbol{v} = \boldsymbol{0}, \quad (\boldsymbol{A} - \lambda\boldsymbol{I}_2)\boldsymbol{w} = \boldsymbol{v} \quad \Rightarrow \quad (\boldsymbol{A} - \lambda\boldsymbol{I}_2)^2\boldsymbol{w} = \boldsymbol{0}.$$

We solve $\boldsymbol{w}$ and then obtain $\boldsymbol{v}$ via $\boldsymbol{v} = (\boldsymbol{A} - \lambda\boldsymbol{I}_2)\boldsymbol{w}$. More precisely, we have

$$\begin{aligned} &(\boldsymbol{A} - \lambda\boldsymbol{I}_2)^2\boldsymbol{w} = \boldsymbol{0} \quad \Rightarrow \quad \begin{pmatrix} -3 & 1 \\ -9 & 3 \end{pmatrix} \begin{pmatrix} -3 & 1 \\ -9 & 3 \end{pmatrix} \begin{pmatrix} w_1 \\ w_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \\ &\Rightarrow \quad \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} w_1 \\ w_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \end{aligned}$$

so $\boldsymbol{w}$ can be any nonzero vector. We take $\boldsymbol{w} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ and compute

$$\boldsymbol{v} = (\boldsymbol{A} - \lambda\boldsymbol{I}_2)\boldsymbol{w} = \begin{pmatrix} -3 & 1 \\ -9 & 3 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \end{pmatrix}.$$

Thus, the second LI solution is

$$\boldsymbol{x}_2(t) = te^{3t} \begin{pmatrix} 1 \\ 3 \end{pmatrix} + e^{3t} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} t \\ 3t + 1 \end{pmatrix} e^{3t}.$$

Then the general solution of the homogeneous system is

$$\boldsymbol{x}_c(t) = c_1 \begin{pmatrix} 1 \\ 3 \end{pmatrix} e^{3t} + c_2 \begin{pmatrix} t \\ 3t + 1 \end{pmatrix} e^{3t} = \begin{pmatrix} e^{3t} & te^{3t} \\ 3e^{3t} & (3t + 1)e^{3t} \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}.$$

We denote the fundamental matrix by $\boldsymbol{X}$. We look for a particular solution of the form

$$\boldsymbol{x}_p(t) = \boldsymbol{X}\boldsymbol{u} \quad \Rightarrow \quad \boldsymbol{X}\boldsymbol{u}' = \boldsymbol{F} \quad \Rightarrow \quad \boldsymbol{u}' = \boldsymbol{X}^{-1}\boldsymbol{F}.$$

We find readily that

$$\boldsymbol{X}^{-1}\boldsymbol{F} = e^{-3t}\begin{pmatrix} 3t+1 & -t \\ -3 & 1 \end{pmatrix}\begin{pmatrix} 0 \\ t \end{pmatrix} = e^{-3t}\begin{pmatrix} -t^2 \\ t \end{pmatrix}$$

and

$$\boldsymbol{u} = \int \boldsymbol{X}^{-1}\boldsymbol{F}\,dt = \begin{pmatrix} -\int t^2 e^{-3t}dt \\ \int te^{-3t}dt \end{pmatrix} = e^{-3t}\begin{pmatrix} \frac{1}{3}t^2 + \frac{2}{9}t + \frac{2}{27} \\ -\frac{1}{3}t - \frac{1}{9} \end{pmatrix}.$$

Then we have

$$\boldsymbol{x}_p(t) = \boldsymbol{X}\boldsymbol{u} = \begin{pmatrix} 1 & t \\ 3 & 3t+1 \end{pmatrix}\begin{pmatrix} \frac{1}{3}t^2 + \frac{2}{9}t + \frac{2}{27} \\ -\frac{1}{3}t - \frac{1}{9} \end{pmatrix} = \begin{pmatrix} \frac{1}{9}t + \frac{2}{27} \\ \frac{1}{9} \end{pmatrix}.$$

Finally, we obtain the general solution

$$\boldsymbol{x}(t) = \frac{1}{9}\begin{pmatrix} t + \frac{2}{3} \\ 1 \end{pmatrix} + c_1 e^{3t}\begin{pmatrix} 1 \\ 3 \end{pmatrix} + c_2 e^{3t}\begin{pmatrix} t \\ 3t+1 \end{pmatrix}.$$

5. Solve the system

$$\begin{aligned} x''(t) &= -2x'(t) - 5y(t) + 3 \\ y'(t) &= x'(t) + 2y(t) \\ x(0) &= 0, \quad x'(0) = 0, \quad y(0) = 1 \end{aligned}$$

by reformulating it as a system of first-order equations.

Solution: We introduce the variables

$$x_1 = x, \quad x_2 = x', \quad x_3 = y.$$

We can then rewrite the above system as

$$\begin{cases} x_1' = x_2, \\ x_2' = -2x_2 - 5x_3 + 3, \\ x_3' = x_2 + 2x_3, \end{cases}$$

with the initial values: $x_1(0) = 0, x_2(0) = 0$ and $x_3(0) = 1$.

We further re-write the equation of (x_2, x_3) in matrix form:

$$\frac{d}{dt}\begin{pmatrix} x_2 \\ x_3 \end{pmatrix}(t) = \begin{pmatrix} -2 & -5 \\ 1 & 2 \end{pmatrix}\begin{pmatrix} x_2 \\ x_3 \end{pmatrix}(t) + \begin{pmatrix} 3 \\ 0 \end{pmatrix}.$$

Denote by $\boldsymbol{A}$ the coefficient matrix. We first find its eigenvalues and eigenvectors. Solve

$$0 = \det(\boldsymbol{A} - \lambda \boldsymbol{I}_2) = \begin{vmatrix} -2-\lambda & -5 \\ 1 & 2-\lambda \end{vmatrix} = (\lambda^2 + 1).$$

Thus, its eigenvalues are $\lambda = \pm \mathrm{i}$. We now look for the corresponding eigenvectors.

For $\lambda = \mathrm{i}$, we solve

$$(\boldsymbol{A} - i\boldsymbol{I})\mathbf{v} = 0$$
$$\begin{pmatrix} -2-i & -5 \\ 1 & 2-i \end{pmatrix}\begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Thus, the eigenvector is:

$$\mathbf{v}_1 = \begin{pmatrix} -2+i \\ 1 \end{pmatrix}$$

For $\lambda_2 = -i$, the calculations are similar, and the eigenvector is:

$$\mathbf{v}_2 = \begin{pmatrix} -2-i \\ 1 \end{pmatrix}$$

So fundamental solution matrix, $\boldsymbol{X} = \begin{pmatrix} -2\cos t - \sin t & -2\sin t + \cos t \\ \cos t & \sin t \end{pmatrix}$.

By variation of parameters, particular solution for system of (x_2, x_3), $\boldsymbol{x}_p = \begin{pmatrix} -6 \\ 3 \end{pmatrix}$.

Now we apply the initial condition $x_2(0) = 0$ and $x_3(0) = 1$, we get,

$x_2(t) = -2(-2\cos t - \sin t) + 2(-2\sin t + \cos t) - 6$

$x_3(t) = -2\cos t + 2\sin t + 3$

So, with $x_0(0) = 0$, $x_1 = \int_0^t x_2(s)ds = 6\sin t + 2\cos t - 6t - 2$.

Finally,

$$\begin{cases} x(t) = 6\sin t + 2\cos t - 6t - 2 \\ y(t) = -2\cos t + 2\sin t + 3 \end{cases} \tag{2}$$